

Topical Antifungal Prescribing for Medicare Part D Beneficiaries — United States, 2021

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Abstract

Incorrect use of topical antifungals and antifungal-corticosteroid combinations is likely contributing to the global emergence and spread of severe antimicrobial-resistant superficial fungal infections, which have recently been detected in the United States. Understanding prescribing patterns is an initial step in establishing and promoting recommended use of these medications. Using 2021 Medicare Part D data, CDC examined prescription volumes, rates, and costs for topical antifungals (including topical combination antifungal-corticosteroid medications). Total prescription volumes were compared between higher-volume prescribers (top 10% of topical antifungal prescribers by volume) and lower-volume prescribers. During 2021, approximately 6.5 million topical antifungal prescriptions were filled (134 prescriptions per 1,000 beneficiaries), at a total cost of \$231 million. Among 1,017,417 unique prescribers, 130,637 (12.8%) prescribed topical antifungals. Primary care physicians wrote the highest percentage of prescriptions (40.0%), followed by nurse practitioners or physician assistants (21.4%), dermatologists (17.6%), and podiatrists (14.1%). Higher-volume prescribers wrote 44.2% (2.9 million) of all prescriptions. This study found that enough topical antifungal prescriptions were written for approximately one of every eight Medicare Part D beneficiaries in 2021, and 10% of antifungal prescribers prescribed nearly one half of these medications. In the setting of emerging antimicrobial resistance, these findings highlight the importance of expanding efforts to understand current prescribing practices while encouraging judicious prescribing by clinicians and providing patient education about proper use.

Introduction

Superficial fungal skin infections have an estimated lifetime prevalence of more than 20% worldwide and are particularly common among adults aged ≥65 years (1–3). The emergence

and spread of antimicrobial-resistant superficial fungal infections, especially dermatophytosis (also known as ringworm or tinea), has led to large outbreaks of extensive, recalcitrant skin infections in South Asia that frequently do not respond to topical antifungals or first-line oral therapies. This emergence and spread are likely exacerbated by the overuse and misuse of topical antifungals, particularly antifungal-corticosteroid combination creams (1,4). Cases of antimicrobial-resistant dermatophytosis have been identified in at least 11 U.S. states (5), with patients experiencing extensive lesions and delays in diagnosis (6). In the United States, nonrecommended topical antifungal prescribing is likely common, because diagnosis of cutaneous fungal infections by visual inspection is frequently incorrect, including among board-certified dermatologists (7), and clinicians across

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specialties rarely perform confirmatory diagnostic testing (2,8). Understanding prescribing patterns, including identification of clinicians who prescribe a disproportionate volume of topical antifungals, might help establish and promote correct use of these medications (9). Centers for Medicare & Medicaid Services (CMS) data were used to characterize prescribing volume of topical antifungal medications among Medicare Part D beneficiaries in the United States during 2021.

Methods

Data Source

Approximately 48.8 million (76%) Medicare beneficiaries were enrolled in the Part D prescription drug benefit program in 2021,^{*,†} the most recent year of data available for this study. The publicly available CMS Medicare Part D Prescribers-by-Provider and Drug data set[§] contains information on the total number of prescriptions (including refills) and total drug

costs,[¶] aggregated by National Provider Identifier number and drug name. The data set excludes clinician records with <11 prescriptions. Prescribers from outside the United States or whose U.S. Census Bureau region was unknown were excluded from the analysis.^{**} The Medicare Monthly Enrollment dataset^{††} was used to ascertain the total number of Medicare Part D beneficiaries.

Data Analysis

Prescription volume (i.e., number of prescriptions), total costs, and average costs for topical antifungal and topical antifungal-corticosteroid drugs covered by Medicare Part D^{§§} were assessed. Prescription rates per 1,000 beneficiaries were calculated by drug and region and to calculate prescription rates per 1,000 beneficiaries by drug and region. Prescription rates per prescriber were calculated by tabulating number of providers overall and by provider type (primary care physician [internal medicine or family medicine physician], nurse

*Medicare, the federal health insurance program, is available to most persons aged ≥65 years, certain persons aged <65 years who are receiving disability benefits, and persons with end stage renal disease. Medicare Part D is the part of Medicare that helps cover the cost of prescription drugs. <https://www.medicare.gov/what-medicare-covers/your-medicare-coverage-choices/whats-medicare>

† <https://data.cms.gov/summary-statistics-on-beneficiary-enrollment/medicare-and-medicare-reports/cms-program-statistics-medicare-part-d-enrollment>

§ <https://data.cms.gov/provider-summary-by-type-of-service/medicare-part-d-prescribers/medicare-part-d-prescribers-by-provider-and-drug>

¶ Cost includes the ingredient cost, dispensing fee, and sales tax and is based on the amounts paid by the Part D plan, Medicare beneficiary, government subsidies, and any other third-party payers. <https://data.cms.gov/resources/medicare-part-d-prescribers-by-provider-and-drug-data-dictionary>

** https://www2.census.gov/geo/pdfs/maps-data/maps/reference/us_regdiv.pdf

†† <https://data.cms.gov/summary-statistics-on-beneficiary-enrollment/medicare-and-medicare-reports/medicare-monthly-enrollment>

§§ Butenafine, butoconazole, ciclopirox, clotrimazole, clotrimazole-betamethasone dipropionate, econazole, efinaconazole, ketoconazole, luliconazole, miconazole, naftifine, nystatin, nystatin-triamcinalone, oxiconazole, sertaconazole, sulconazole, tavaborole, and terconazole.

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practitioner or physician assistant, dermatologist, podiatrist, and other) in the Medicare Part D Prescribers—by Provider and Drug data set as the denominator.

Higher-volume prescribers were defined as those within the top 10th percentile of prescriber-level topical antifungal prescriptions by volume. The volume and percentage of topical antifungal prescriptions written by higher-volume prescribers compared with lower-volume prescribers (bottom 90th percentile) were assessed overall and by prescriber type. Analyses were conducted using SAS software (version 9.4; SAS Institute). This activity was reviewed by CDC, deemed not research, and was conducted consistent with applicable federal law and CDC policy.^{¶¶}

Results

Prescriptions

During 2021, a total of 6.5 million topical antifungal prescriptions were filled by Part D beneficiaries (overall rate = 134.0 prescriptions per 1,000 beneficiaries) (Table 1). By volume, the most common prescriptions were for ketoconazole (2.4 million [36.6%]), nystatin (1.9 million [29.0%]), and clotrimazole-betamethasone dipropionate (0.9 million [14.7%]). The total cost for all topical antifungal prescriptions was \$231 million. The highest average per prescription costs

were for efinaconazole (\$1,035.38), tavaborole (\$784.63), and oxiconazole (\$729.84); the lowest average costs were for nystatin (\$25.66), clotrimazole-betamethasone dipropionate (\$27.82), clotrimazole (\$30.36), and ketoconazole (\$30.69). By U.S. Census Bureau region, the highest prescription rate was in the Northeast (188.0 prescriptions per 1,000 beneficiaries), followed by the South (138.1 per 1,000).

Prescribers

Among 1,017,417 unique prescribers, 130,637 (12.8%) prescribed topical antifungals (Table 2). The number of prescriptions per provider was highest for dermatologists (87.1), followed by podiatrists (67.2), and primary care physicians (12.3). Among 6.5 million topical antifungal prescriptions, the most were written by primary care physicians (2.6 million [40.0%]), followed by nurse practitioners or physician assistants (1.4 million [21.4%]), dermatologists (1.1 million [17.6%]), and podiatrists (0.9 million [14.1%]). Among all topical antifungal prescriptions, 44.2% (2.9 million) were written by the top 10% (13,106) of topical antifungal prescribers. By provider type, the percentage of topical antifungal prescriptions written by higher-volume topical antifungal prescribers ranged from 35.5% for dermatologists to 54.8% for podiatrists.

Discussion

This analysis of publicly available CMS data found that 6.5 million topical antifungal prescriptions (enough to provide

^{¶¶} 45 C.F.R. part 46.102(l)(2), 21 C.F.R. part 56; 42 U.S.C. Sect. 241(d); 5 U.S.C. Sect. 552a; 44 U.S.C. Sect. 3501 et seq.

TABLE 1. Number of topical antifungal prescriptions, prescriptions per 1,000 beneficiaries, and cost, by drug and U.S. Census Bureau region — United States, 2021

Characteristic	No. of prescriptions (% of all prescriptions)	Prescriptions per 1,000 beneficiaries	Aggregate cost, all prescriptions, USD*	Avg. cost per prescription, USD*
Antifungal drug				
Ketoconazole [†]	2,364,169 (36.6)	49.1	72,556,081.61	30.69
Nystatin	1,871,368 (29.0)	38.9	48,025,095.10	25.66
Clotrimazole-betamethasone dipropionate	945,838 (14.7)	19.6	26,311,901.12	27.82
Ciclopirox	657,986 (10.2)	13.7	22,739,088.92	34.56
Clotrimazole	397,603 (6.2)	8.3	12,070,066.47	30.36
Econazole	75,675 (1.2)	1.6	4,414,180.46	58.33
Nystatin-triamcinolone	55,276 (0.9)	1.1	2,928,976.33	52.99
Terconazole	32,203 (0.5)	0.7	1,051,091.05	32.64
Efinaconazole	17,881 (0.3)	0.4	18,513,585.90	1,035.38
Oxiconazole	14,892 (0.2)	0.3	10,868,838.98	729.84
Naftifine	13,532 (0.2)	0.3	5,018,673.14	370.87
Tavaborole	8,317 (0.1)	0.2	6,525,735.61	784.63
Other [§]	400 (—)	—	185,439.42	463.60
U.S. Census Bureau region				
Northeast	1,677,727 (26)	188.0	78,235,437.54	46.63
Midwest	1,198,093 (19)	112.2	34,204,692.89	28.55
South	2,489,321 (39)	138.1	74,554,713.01	29.95
West	1,089,999 (17)	103.5	44,213,910.67	40.56
Total	6,455,140 (100)	134.0	231,208,754.11	35.82

* All costs are in U.S. dollars.

[†] It was not possible to distinguish oral from topical ketoconazole, but ketoconazole was included in the analysis because oral ketoconazole use is discouraged because of safety concerns. <https://www.fda.gov/drugs/drug-safety-and-availability/fda-drug-safety-communication-fda-warns-prescribing-nizoral-ketoconazole-oral-tablets-unapproved>

[§] Butenafine (41 prescriptions), luliconazole (169), miconazole (155), and sertaconazole (35).

TABLE 2. Number of topical antifungal prescriptions per provider and prescribing volume among higher-volume* and lower-volume† prescribers, by provider type for Medicare Part D beneficiaries — United States, 2021

Provider type (no.)	No. of prescriptions per provider	All topical antifungal prescribers		Higher-volume prescribers (top 10%)		Lower-volume prescribers (bottom 90%)	
		No. of prescribers	No. of prescriptions (% of total) [§]	No. of prescribers	No. of prescriptions (% of total) [¶]	No. of prescribers	No. of prescriptions (% of total) [¶]
Primary care physician (209,169)	12.3	61,735	2,579,045 (40.0)	6,200	973,824 (37.8)	55,535	1,605,221 (62.2)
NP or PA (263,999)	5.2	34,476	1,379,981 (21.4)	3,428	570,577 (41.3)	31,048	809,404 (58.7)
Dermatologist (13,029)	87.1	10,735	1,134,347 (17.6)	1,068	402,750 (35.5)	9,667	731,597 (64.5)
Podiatrist (13,527)	67.2	8,401	909,569 (14.1)	838	498,210 (54.8)	7,563	411,359 (45.2)
Other (517,693)	0.9	15,290	452,198 (7.0)	1,507	165,742 (36.7)	13,783	286,456 (63.3)
Total (1,017,417)	6.3	130,637	6,455,140 (100.0)	13,106	2,851,394 (44.2)	117,531	3,603,746 (55.8)

Abbreviations: NP = nurse practitioner; PA = physician assistant.

* Top 10% of topical antifungal providers, by volume.

† Bottom 90% of topical antifungal providers, by volume.

§ Column percentage.

¶ Row percentage.

one prescription to more than one eighth of all beneficiaries) were written for Medicare Part D beneficiaries in 2021, at a cost of \$231 million. The actual volume of topical antifungal use among the study population is likely considerably higher than that identified in this study because most topical antifungals can be purchased over the counter without a prescription; such topical antifungal use is not recorded in CMS data and is an important consideration for potential antifungal stewardship efforts. The large volume of topical antifungals used in the United States warrants increased attention given the infrequent use of confirmatory testing, inaccuracy of diagnosis made by physical examination alone, and the recent emergence of severe and antimicrobial-resistant superficial skin infections (1,5,6). To help control the emergence and spread of antimicrobial-resistant superficial fungal infections and help promote the appropriateness of topical antifungal prescribing, health care providers could use diagnostic testing*** whenever possible to confirm suspected superficial fungal infections. Further, health care providers can educate patients about prognosis, benefits, and harms of topical antifungal and combination antifungal-corticosteroid treatment (both prescription and over-the-counter), and the importance of using these medications as prescribed or according to manufacturer instructions.

Variation in Prescribing by Prescriber Type and Region

The largest number of topical antifungal prescriptions was written by primary care physicians, nurse practitioners, or physician assistants, suggesting that efforts to determine and improve appropriateness of prescribing could prioritize these groups. Although dermatologists and podiatrists had lower prescribing volumes compared with other groups, they had higher per-provider prescribing rates. This observation could reflect that dermatologists and podiatrists might see patients

with superficial fungal infections more frequently than do other provider types. In contrast to systemic antibiotic prescribing, which is highest in the South (9), topical antifungal prescribing rates were highest in the Northeast. Reasons for this finding are unclear but could reflect a higher prevalence of superficial fungal infections, more ready access to medical care, or less frequent use of over-the-counter topical antifungal medications in the Northeast compared with that in other regions.

High Volume Prescribers and Prescriptions

As with antibiotic prescribing for Medicare Part D beneficiaries, 10% of prescribers wrote a disproportionately large percentage (>40%) of topical antifungal prescriptions (9). Among podiatrists, the top 10% of prescribers wrote more than one half of topical antifungal prescriptions. These findings suggest potential opportunities to prioritize higher-volume topical antifungal prescribers for antimicrobial stewardship interventions using evidence-based techniques such as peer comparison audit and feedback; however, additional data are needed to determine whether topical antifungal prescribing rates correlate with rates of incorrect prescribing, as shown for systemic antibiotics in primary care settings (9).

The large volume of clotrimazole-betamethasone dipropionate prescriptions (0.9 million; 15% of all topical antifungal prescriptions) is potentially concerning, as use of combination topical medications containing corticosteroids and antifungal agents has been proposed as a potential driver of emerging antimicrobial-resistant dermatophytosis (10). In addition, clotrimazole-betamethasone dipropionate contains a high-potency steroid that can cause skin damage if applied to intertriginous areas as well as hypothalamic-pituitary-adrenal axis suppression if used for a prolonged time or over a large body surface area.††† Clinicians should be aware of the potential risks

*** <https://www.cdc.gov/fungal/diseases/ringworm/health-professionals.html>

††† https://www.accessdata.fda.gov/drugsatfda_docs/label/2014/018827s046lbl.pdf

Summary

What is already known about this topic?

Severe antimicrobial-resistant superficial fungal infections have recently been detected in the United States; evaluating topical antifungal use is an initial step in developing strategies to prevent the global emergence and spread of these infections.

What is added by this report?

A total of 6.5 million topical antifungal prescriptions, costing \$231 million, were filled for Medicare Part D beneficiaries in 2021, approximately one prescription for every eight beneficiaries. Most prescriptions were written by primary care physicians, nurse practitioners, or physician assistants.

What are the implications for public health practice?

The large volume of topical antifungal prescriptions in the context of emerging resistance highlights the need to better understand current prescribing practices and to encourage judicious prescribing by clinicians and improve patient education about recommended use.

associated with clotrimazole-betamethasone dipropionate use and consider alternatives such as antifungal monotherapy, with a short course of low-potency corticosteroid treatment added if needed for symptoms such as severe pruritis. §§§

Limitations

The findings in this report are subject to at least four limitations. First, the data set does not contain information on individual patients, drug indication (i.e., candidiasis versus dermatophytosis), or diagnostic testing, so prescribing appropriateness could not be determined. Second, the data set analyzed only identifies prescriptions for Medicare Part D beneficiaries and therefore does not represent all Medicare beneficiaries; topical antifungal prescribing patterns might differ among other populations. Third, the data set only contained information on prescription topical antifungals and did not capture over-the-counter topical antifungal use; therefore, actual topical antifungal use is likely underestimated. Finally, this study likely underestimates the total volume of topical antifungal drug prescribing among Medicare Part D beneficiaries because records for some lower-volume prescribers (those with <11 prescriptions per year for any given drug) are not included in the data set, and prescribers whose census region was unknown were excluded.

Implications for Public Health Practice

The substantial volume of topical antifungal and antifungal-corticosteroid prescriptions among Medicare Part D beneficiaries in the setting of emerging resistant infections underscores

the need to evaluate current practices of topical antifungal use. Health care providers should be judicious in prescribing topical antifungals and combination antifungal-corticosteroid medications for suspected superficial fungal infections, using testing when feasible to confirm diagnoses, and can educate patients about the correct use of topical antifungals and combination antifungal-corticosteroids.

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§§§ <https://www.aafp.org/pubs/afp/collections/choosing-wisely/471.html>

Advisory Committee on Immunization Practices Recommended Immunization Schedule for Children and Adolescents Aged 18 Years or Younger — United States, 2024

A. Patricia Wodi, MD¹; Neil Murthy, MD¹; Veronica V. McNally, JD²; Matthew F. Daley, MD³; Sybil Cineas, MD⁴

At its October 2023 meeting, the Advisory Committee on Immunization Practices* (ACIP) approved the Recommended Child and Adolescent Immunization Schedule for Ages 18 Years or Younger, United States, 2024. The child and adolescent immunization schedule, which can be found on the CDC immunization schedule website (<https://www.cdc.gov/vaccines/schedules>), is published annually to consolidate and summarize updates to ACIP recommendations on the vaccination of children and adolescents and to assist health care providers in implementing current ACIP recommendations. The 2024 immunization schedule includes several changes to the cover page, tables, notes, and appendix from the 2023 immunization schedule.[†] In addition, the 2024 child and adolescent immunization schedule includes a new addendum section to summarize new or updated ACIP recommendations that will occur before the next annual update to the child and adolescent immunization schedule. Health care providers are advised to use the cover page, tables, notes, appendix, and addendum together to identify the recommended immunizations for patient populations.

The 2024 child and adolescent immunization schedule is recommended by ACIP (<https://www.cdc.gov/vaccines/acip>) and approved by CDC (<https://www.cdc.gov>), the American Academy of Pediatrics (<https://www.aap.org>), the American Academy of Family Physicians (<https://www.aafp.org/home.html>), the American College of Obstetricians and Gynecologists (<https://www.acog.org/>), the American College of Nurse-Midwives (<https://www.midwife.org>), the American Academy of Physician Associates (<https://www.aapa.org>),

and the National Association of Pediatric Nurse Practitioners (<https://www.napnap.org>).

ACIP's recommendations for the use of each vaccine and other immunizing agents are developed after in-depth reviews of product-related data, including the epidemiology and societal impacts of the vaccine-preventable disease, efficacy and effectiveness of the vaccine or other immunizing agent, safety of the vaccine or other immunizing agent, quality of evidence, feasibility of program implementation, impact on health equity, and economic analyses of immunization policy (1,2). Health care providers should be aware that changes in recommendations for specific vaccines and related agents occur between these annual updates to the child and adolescent immunization schedule.[§] Such changes will be summarized in the new addendum section; however, health care providers are encouraged to refer to ACIP vaccine recommendations for detailed guidance on the use of each product (<https://www.cdc.gov/vaccines/hcp/acip-recs>). An online version of the 2024 child and adolescent immunization schedule and instructions for downloading the schedule app are available on the immunization schedule website (<https://www.cdc.gov/vaccines/schedules>). The use of trade names in the child and adolescent immunization schedule and in this report is for identification purposes only and does not imply endorsement by ACIP or CDC.

Changes in the 2024 Child and Adolescent Immunization Schedule

Changes to the recommendations for vaccines and related agents in the 2024 immunization schedule for children and adolescents aged ≤18 years include new or updated recommendations for influenza vaccine (3), pneumococcal vaccines (4), respiratory syncytial virus monoclonal antibody

* Recommendations for routine immunization of children and adolescents are developed by ACIP, a federal advisory committee chartered to provide expert external advice and guidance to the CDC director on use of vaccines and related agents for the control of vaccine-preventable diseases in the civilian population of the United States. Recommendations for routine immunization of children and adolescents are harmonized to the greatest extent possible with recommendations made by the American Academy of Pediatrics, the American Academy of Family Physicians, the American College of Obstetricians and Gynecologists, the American College of Nurse-Midwives, the American Academy of Physician Associates, and the National Association of Pediatric Nurse Practitioners. ACIP recommendations become official agency guidelines once the recommendation has been adopted by the CDC Director. Additional information about ACIP is available at <https://www.cdc.gov/vaccines/acip>.

[†] Past immunization schedules are available at <https://www.cdc.gov/vaccines/schedules/hcp/schedule-related-resources.html>.

[§] CDC encourages organizations to use syndication as a more reliable method for displaying the most current and accurate immunization schedules on an organization's website rather than copying these schedules to their websites. Use of content syndication requires a one-time step that ensures an organization's website displays current schedules as soon as they are published or revised; instructions for syndication code are available on CDC's website (<https://www.cdc.gov/vaccines/schedules/resource-library/syndicate.html>). CDC also offers technical assistance for implementing this form of content syndication (requests can be emailed to ncirdwebteam@cdc.gov).

(RSV-mAb) (5), respiratory syncytial virus vaccines (RSV) (6), COVID-19 vaccines (7), inactivated poliovirus vaccine (IPV) (8), Mpox vaccine (Mpox) (<https://www.cdc.gov/vaccines/acip/meetings/downloads/slides-2023-10-25-26/04-MPOX-Rao-508.pdf>), and meningococcal serogroups A, B, C, W, Y vaccine (MenACWY-TT/MenB-FHbp) (<https://www.cdc.gov/vaccines/acip/recommendations.html>). Diphtheria and tetanus toxoid adsorbed vaccine (DT), 13-valent pneumococcal conjugate vaccine (PCV13), bivalent COVID-19 mRNA vaccines, and meningococcal serogroups A, C, W, Y polysaccharide diphtheria toxoid conjugate vaccine (MenACWY-D, Menactra) were deleted from all sections of the schedule, because these products are no longer distributed or recommended for use in children and adolescents in the United States.

Other changes include clarification of the recommendations for diphtheria, tetanus, and acellular pertussis vaccine (DTaP), *Haemophilus influenzae* type b vaccine (Hib), human papillomavirus vaccine (HPV), measles, mumps, and rubella vaccine (MMR), serogroup B meningococcal vaccine (MenB), and tetanus, diphtheria, and acellular pertussis vaccine (Tdap). Substantial revisions were made to Table 3, which outlines the immunization schedule by medical indication. The definitions for the legend colors were revised to better highlight additional vaccination recommendations for each medical condition and to harmonize with the adult immunization schedule. Finally, a new addendum section was added, which will list new and updated ACIP recommendations that occur before the next annual update to the child and adolescent immunization schedule.

Cover page

- In the table of abbreviations and trade names, the column header was changed from “vaccine” to “vaccines and other immunizing agents” to account for the inclusion of the newly licensed RSV monoclonal antibody (nirsevimab).
- A sixth step in the “How to Use the Child and Adolescent Immunization Schedule” box was added directing health care providers to review the new Addendum section that lists new or updated ACIP recommendations that occur before the next annual update of the child and adolescent immunization schedule.
- 20-valent pneumococcal conjugate vaccine (PCV20), RSV-mAb (nirsevimab), RSV for maternal vaccination (Abrysvo), Mpox (Jynneos), and pentavalent meningococcal vaccine (MenACWY-TT/MenB-FHbp, [Penbraya]) have been added to the table listing abbreviations and trade names of vaccines and other immunizing agents.
- Diphtheria and Tetanus Toxoid Adsorbed vaccine (DT), 13-valent pneumococcal conjugate vaccine (PCV13), MenACWY-D (Menactra), and bivalent mRNA COVID-19 vaccines were removed from the table listing

abbreviations and trade names of vaccines and other immunizing agents, because they are no longer distributed or recommended for use in the United States.

Table 1 (Routine Immunization Schedule)

- The column header was changed from “vaccine” to “vaccines and other immunizing agents” to account for the inclusion of the newly licensed RSV monoclonal antibody (nirsevimab).
- **COVID-19 row:** The text overlay was revised to reflect updated vaccination recommendations. This text overlay now states, “1 or more doses of updated (2023–2024 Formula) vaccine.”
- **MenACWY row:** Menactra has been deleted.
- **Mpox row:** A new row was added for Jynneos with the column for age 18 years highlighted in purple reflecting the risk-based recommendation for this age group.
- **Pneumococcal conjugate row:** PCV20 has been added and PCV13 has been deleted.
- **Pneumococcal polysaccharide vaccine (PPSV23) row:** This row has been deleted because PPSV23 is no longer routinely recommended for all children and adolescents aged ≥ 2 years at increased risk for invasive pneumococcal disease. It is still recommended in certain circumstances.
- **RSV-mAb row:** A new row has been added with the columns for ages birth–7 months highlighted in yellow to indicate the recommended age for routine immunization. The overlaying text, “1 dose depending on maternal RSV vaccination status” was also added. In addition, age 8–19 months is highlighted in purple to reflect the risk-based recommendation for this age group.
- **RSV row:** A new row was added for Abrysvo (Pfizer Inc.) and ages 11–18 years are highlighted in purple with the overlaying text, “Seasonal administration during pregnancy” added to reflect the recommendation for the use of Abrysvo (Pfizer Inc.) during pregnancy.

Table 2 (Catch-up Immunization Schedule)

- **DTaP row:** Language for the minimum interval between doses 4 and 5 was added to clarify when a fifth dose is indicated. The text reads, “A fifth dose is not necessary if the fourth dose was administered at age ≥ 4 years and ≥ 6 months after dose 3.”
- **MenACWY row:** Menactra has been deleted.

Table 3 (Immunization by Medical Indication Schedule)

- A sentence was added to the header of Table 3 stating that medical conditions are often not mutually exclusive and that health care providers should review all relevant columns in the Table if multiple conditions are present.

- The column header was changed from “vaccine” to “vaccines and other immunizing agents” to account for the inclusion of the newly licensed RSV monoclonal antibody (nirsevimab).
- **Legend:** The definitions of the yellow, purple, and gray colors boxes in the legend were revised. Based on the revised definitions, the colors for many of the rows in this table have changed. In addition, the checked yellow color was changed to a brown color to harmonize with the 2024 adult immunization schedule.
- **Mpox row:** A new row was added for Jynneos. Across all medical indications listed, the entire row is purple reflecting the risk-based recommendation for Mpox vaccination. In the pregnancy column, an overlaying text, “See Notes” has been added, directing health care providers to review the pregnancy bullet in the Mpox vaccination notes.
- **RSV-mAb row:** A new row was added to summarize nirsevimab immunization recommendations by medical condition. The columns for both immunocompromised status (excluding HIV infection) and HIV infection with CD4 <15% or <200 cells per mm³ is highlighted in brown and an overlaying text “2nd RSV season” was added. In addition, the column for heart disease or chronic lung disease is also highlighted in brown with the overlaying text “2nd RSV season for chronic lung disease.”
- **RSV row:** A new row was added for use of Abrysvo (Pfizer Inc.) during 32–36 weeks’ gestation. The pregnancy column is highlighted in yellow with overlaying text of “seasonal administration” added to indicate that the maternal RSV vaccination recommendation is on the basis of RSV seasonality.

Vaccine Notes

The notes for each vaccine and related agent are presented in alphabetical order. Edits have been made throughout the Notes section to harmonize language, to the greatest extent possible, with that in the adult immunization schedule.

- **Additional information:** The text for vaccine injury compensation was revised to add Mpox and RSV to the list of vaccines not covered by the National Vaccine Injury Compensation Program. Mpox is covered by the Countermeasures Injury Compensation Program.
- **COVID-19:** The language in the “Routine vaccination” and “Special situations” sections was revised to reflect the current COVID-19 vaccination recommendations for children and adolescents. The number of doses needed and intervals between doses might vary on the basis of a patient’s previous vaccination history, immunocompromised status, and the vaccine product used. The “Routine vaccination” section describes the recommendations for

the general population, and the “Special situations” section describes the recommendations for persons who are moderately or severely immunocompromised. In addition, hyperlinks to the current COVID-19 vaccination schedules as well as Emergency Use Authorization indications for COVID-19 vaccines are included.

- **DTaP:** Language in the “Routine vaccination” section was revised to clarify primary and booster doses.
- **HPV:** In the “Routine vaccination” section, the recommendation for interrupted schedules was removed because that information is also presented on the Cover Page and applicable to all vaccines. In addition, to improve clarity, the words, “of any valency” were added to the bullet, “No additional dose recommended when any HPV vaccine series *of any valency* has been completed using the recommended dosing intervals.”
- **Influenza:** A hyperlink to the 2023–24 influenza recommendations and a bullet for the 2024–25 influenza recommendations were added. In the “Special situations” section, all bullets describing recommendations for persons with a history of egg allergy were removed. Persons with a history of egg allergy of any severity can be vaccinated with any influenza vaccine indicated for the recipient’s age and health status, with no additional safety considerations. A note describing this recommendation was added at the end of the “Special situations” section.
- **MMR:** The bullet, “If MMRV is used, the minimum interval between MMRV doses is 3 months” was moved to the end of the notes section. In addition, the “Routine vaccination,” “Catch-up vaccination,” and “Special situations” sections were revised to clarify that this minimal interval is applicable to all sections.
- **MenACWY:** All reference to Menactra was removed because this vaccine is no longer distributed in the United States, and any remaining doses of this product expired in October 2023. In addition, information about the use of the newly licensed pentavalent meningococcal vaccine (Penbraya) is included at the end of the MenACWY notes.
- **MenB:** A note summarizing recommendations for Penbraya was added. In addition, a link to a resource to assist health care providers with shared clinical decision-making recommendations for MenB vaccination was added.
- **Mpox:** A new section describing the recommendations for use of Jynneos in adolescents aged 18 years, including sexual risk factors and vaccination during pregnancy, was added.
- **Pneumococcal:** The “Routine vaccination,” “Catch-up vaccination,” and “Special situations” sections have been updated with the new recommendations for use of 15-valent pneumococcal conjugate vaccine (PCV15), PCV20, and PPSV23. PCV13 was deleted from all